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Project Proposal

PRIOT Smart Track: Pilot Implementation for Dolomiti NordicSki Val Casies / Gsiesertal, Season 2026/27

1. Context & Goals

Nordic skiing presents a structural challenge that alpine ticketing models cannot address: trails are freely accessible, geographically dispersed, and impossible to gate. Dolomiti NordicSki (DNS), a cross-border network of eight Nordic ski regions spanning South Tyrol, the Veneto, and Austria, has identified this as the central problem to solve. The strategic answer, as articulated in the DNS internal concept, is a paradigm shift: away from blanket access control and toward an intelligent, adaptive, data-driven system with minimal complexity and maximum scalability.

PRIOT Smart Track is named in the DNS system concept as the technological foundation of the DNS ticketing architecture. This proposal details the first concrete step: a contained, single-region pilot in Val Casies / Gsiesertal during the 2026/27 winter season.

Phase 1: Pilot (this proposal): The pilot operates with primarily daily tickets in a single region (we propose the use of testing seasonally with BETA users as well). Day tickets are deliberately chosen as the simplest vehicle: no cross-regional clearing, a single point of sale, and a contained data scope, to validate the PRIOT Smart Track stack under real operational and weather conditions before it carries the DNS network season card. The pilot validates three things: the technology (RFID sticker tickets, UHF validation, fixed measurement station), the operational workflow (issuance at the tourist office, mobile validation by inspectors, admin oversight), and user acceptance by Gsiesertal guests and locals.

Phase 2+: Network rollout (indicative, Section 7): A successful pilot unlocks the full DNS vision: the DNS season card (armband, UHF automatic detection, network-wide validity), centralised user data, automated clearing, personalised user services across all regions, and, in time, a fixed measurement-station network across the Alpine Nordic corridor supported by EU funding mechanisms such as Alpine Space Interreg.

Gsiesertal was selected by DNS as the pilot region due to its single management structure, one central point of sale, approximately 40 km of connected trails, reliable snow conditions with early-season opening, a suitable location for a fixed station on the main trail, and strong local commitment to participate actively in the project.

2. Solution: PRIOT Smart Track

PRIOT Smart Track is a cloud-first, integrated hardware and software platform for access management, ticket validation, and usage analytics in Nordic and cross-country ski environments. The system has been operational at the Planica Nordic Centre (Slovenia) since 2022, where it manages daily and seasonal passes across two of the most-used ski jumping and cross-country hills in the world.

The platform consists of four integrated components:

- **Fixed Measurement / Validation Station:** A weatherproof field station with two UHF antennas, LTE connectivity, and 230VAC or solar DC power. Mounted at a strategic location on the trail, it automatically reads RFID-enabled tickets as skiers pass: without requiring the skier to stop or interact with the system. Real-time data is transmitted to the central cloud database. Offline mode is supported: the station buffers events locally and syncs when connectivity is restored.
- **Handheld Inspector Device:** A rugged Android device with an integrated long-range UHF antenna. Inspectors on the trail can validate tickets from 5–7 metres away without stopping the skier. On a valid scan, the device displays a minimal profile (photo, first name, last name, validity status) and clears automatically: no persistent access to the user database. Devices are password-protected; a lost or stolen device presents no data exposure risk.
- **Ticket Issuance Software:** A desktop application (Windows) installed at the point of sale. Staff issue and activate tickets via a USB UHF reader, photograph the guest if required, and confirm validity. Batch issuance of pre-programmed daily-ticket stickers is supported for high-throughput POS scenarios.
- **Admin Dashboard & Analytics:** A web-based administration portal providing real-time overview of active/expired tickets, usage counts, measurement-station status, and frequency analytics (skier passages per hour/day/location). Exportable data for regional reporting. Role-based access ensures each operator sees only their region's data.
- **RFID Daily Ticket (sticker format):** A compact sticker ticket with an embedded UHF RFID chip and a QR code. The sticker can be mounted on a ski pole or jacket, placing it at the correct distance from the body for reliable long-range UHF detection. The QR code serves as a personal profile link (usage history, point collection, incentives in future phases). Each ticket is pre-programmed and loaded into the system at issuance.

3. Pilot Scope & Deliverables: Gsiesertal

3.1 Functional Scope

- Daily ticket issuance at the central tourist office POS (UHF sticker tickets, with QR code)
- Ticket validation via 3 handheld inspector devices (UHF long-range, 5-7 m detection range)
- Mobile validation fallback via QR code scan (smartphone-based, for regions/controllers without a dedicated device)
- 2 fixed measurement/validation station on the main Gsiesertal trail (automatic UHF detection, live passage counting)
- Admin dashboard: real-time ticket status, usage analytics, station monitoring
- Region-scoped configuration: day tickets valid in Gsiesertal only
- Data-minimised validation view on inspector devices (name, photo, valid/invalid: no browsable database)

3.2 Kick-off Workshop + Field inspection

An in-person workshop (PRIOT team + DNS coordination) to be held as soon as the contract is signed. Objectives: confirm functional requirements, agree on data fields, define POS workflow, confirm station placement on the trail map, align on ticket design/graphic specifications, agree on formats and integration points, and plan the deployment calendar.

3.3 Roles & Responsibilities

DNS / Gsiesertal region	PRIOT
Provide POS location and Windows PC with internet access	Supply, configure and install all software components
Provide site access and a suitable post for the fixed station (metal pole, concrete base for transporting the unit)	Supply, deliver and install all hardware (station, validators, USB readers)
Coordinate local operational staff; train staff on issuance workflow	Deliver 5,000 programmed RFID daily tickets to DNS
Define ticket pricing, validity rules, and ticket graphic/branding	On-site setup, go-live testing, and staff training
Collect and report operational feedback during pilot season	Remote monitoring, incident response, and technical support throughout pilot

4. Technical Conditions

Hosting: PRIOT-hosted cloud (EU data residency) for the duration of the pilot. This is the fastest path to deployment, avoids on-premise setup complexity, and allows PRIOT to iterate on any issues rapidly during the season. Data governance and GDPR responsibilities can be addressed during the kickoff workshop.

Data & Privacy (GDPR): All pilot ticket data and customer information remain within a single region. Validation devices display only the minimal required information (photo, name, valid/not valid) and clear the screen on completion of each check. Devices are password-protected. No persistent browsable database access is available to field staff. This design makes the pilot's data governance straightforward and manageable, as noted by DNS.

Hardware prerequisites (DNS/region responsibility): Windows PC at POS with internet access; a suitable mounting post or mast at the agreed station location (PRIOT provides specs); stable LTE coverage at the station site (to be confirmed at the workshop).

5. Timeline

Period	Activity
July 2026	Board decision + contract signing. Kick-off workshop (in-person, PRIOT visits DNS/Gsiesertal). Requirements finalised, station placement confirmed, ticket design agreed.
July–September 2026	Software configuration for Gsiesertal (region scoping, offline mode, ticket parameters). Hardware procurement and pre-testing. UHF tickets manufactured and programmed. Internal QA and staging tests.
October 2026	On-site installation of a fixed measurement station. POS software installation and USB reader setup. Delivery of handheld validators and RFID tickets. Staff training. Go-live testing with real tickets.
November 2026–Season End	Live pilot operation. Remote monitoring and technical support by PRIOT. Operational feedback collection. Usage data and analytics available in real time via admin dashboard.
Post-season 2027	Pilot evaluation. Decision on Phase 2 (DNS season card network rollout) and additional regions joining.

Note: The October go-live target is achievable because the core PRIOT Smart Track software is already developed and operational at Planica. The critical path is the kick-off workshop, which must be confirmed promptly after the board decision to protect this schedule.

6. Pilot Pricing: Val Casies / Gsiesertal

All prices are in EUR, excluding VAT, and apply for the 6-month pilot period (deployment + winter season 2026/27). Training and support are included in the pilot fee at no additional charge. There are no annual operating costs billed separately during the pilot period: the flat fee covers all platform access, hosting, and support for the pilot duration.

#	Category	Line item	Amount (ex-VAT)
1	Platform & software licensing	Pilot software bundle: ticket issuance desktop app, mobile validator app, admin dashboard web app. 6-month license, all platform licenses included. + BONUS: custom DNS personalisation options on my-profile for each ticket (defined on the workshop)	4.500 €
		Real-Time Data Collection with live database: software running on measurement station hardware, live cloud database, real-time sync, 6-month pilot license.	7.000 €
2	RFID tickets	5,000 × daily ticket RFID stickers (embedded UHF chip + QR code, pre-programmed, custom graphic) @ €0.86/pcs	4.300 €

#	Category	Line item	Amount (ex-VAT)
3	Validators & readers	3× handheld validator (Android, integrated long-range UHF antenna, 5–7 m detection range) @ €1,800/pcs	5.400 €
		2× USB UHF office reader (for POS ticket issuance and activation) @ €200/pcs	400 €
4	Fixed measurement station	1× fixed measurement/validation station: 2× UHF antenna, LTE connectivity, 230VAC or solar DC power, weatherproof enclosure, incl. mounting hardware	7.000 €
5	Kick-off workshop + requirements	In-person on-site visit, requirements finalisation, workflow definition, station placement, format/integration agreement	2.500 €
6	On-site setup & system startup	On-site hardware installation, software deployment, POS configuration, staff training, go-live validation	4.500 €
7	Training & support	Included in pilot fee: remote technical support and monitoring throughout the pilot season	0 €
8	Annual operating costs (pilot)	Included in 6-month pilot fee: no separate annual cost during the pilot period	0 €
TOTAL PILOT: ex-VAT			35.600 €

Payment terms: 50% on contract signature (covers hardware procurement and workshop); 50% on go-live (October 2026). Payment due 8 days of invoice. All prices exclude VAT.

Conditions: This offer represents an obligation of effort, not a guarantee of specific outcomes.

7. Indicative Network-Wide Rollout (Phase 2)

The figures and scope in this section are indicative only and intended to support DNS's internal planning and board discussion. They are not contractual commitments. A formal Phase 2 offer will be prepared following pilot evaluation and a joint requirements workshop.

A successful Gsiesertal pilot unlocks the full DNS Smart Track vision as described in the DNS system concept: the DNS season card with UHF/NFC armband as the physical carrier, automatic detection at fixed measurement stations across the network, personalised user accounts (LOCAL / GUEST / PREMIUM), incentivisation mechanics (points, early-bird pricing, challenges), and centralised usage analytics feeding the DNS clearing system.

7.1 Technical Scope: Phase 2

- **DNS season card:** UHF armband (UHF, linked to personal account), network-wide validity, UHF automatic detection at fixed stations in all participating regions. Armband produced by DNS's supplier (SAXE) to PRIOT's buffer specifications for optimal signal quality.
- **Clearing API integration:** PRIOT provides an API endpoint for the DNS web shop / purchasing platform to pass season-card purchase events, activate cards, and feed usage data back for the Hausloipe-based revenue

distribution. This is the data backbone connecting the physical system to DNS's financial clearing.

- **Additional regions:** Each new region joining the network requires local hardware (validators, USB readers, fixed stations as desired) plus region-scoping configuration in the central platform. Onboarding per region is a defined, replicable process.
- **User profile portal:** Web-based personal profile for each cardholder: usage history, kilometres tracked, incentive points, and validity confirmation.
- **Data governance:** PRIOT-hosted cloud (licensed model) based on DNS's GDPR.

7.2 Indicative Hardware Costs per Region

Item	Indicative unit price (ex-VAT)	Notes
Fixed long-range measurement station (UHF antenna ×2, LTE, solar)	~4.500 €	Volume and configuration dependent
Fixed manual validation station (UHF short-range, LTE, solar)	~1.500 €	Volume and configuration dependent
Handheld validator	~1.400 €	Significant discounts at 30+ units network-wide
USB UHF office reader	~200 €	Per POS location
DNS season card (UHF, programmed, printed)	~2,20 € / card	Armband produced separately by DNS/SAXE
UHF daily ticket (UHF, programmed, printed)	~0,70 € / ticket	Volume discount pricing from ~10,000 units

7.3 Annual Maintenance & Software Costs (Phase 2+)

From Phase 2, the pilot flat fee transitions to a commercial model sized against actual network usage (number of regions, active cards, annual ticket volume, validation events). PRIOT will benchmark the required system bandwidth post-pilot and propose a transparent annual fee structure: either as a platform license (cloud model) or an annual maintenance contract. The goal is a predictable, region-level cost that DNS can present to each participating region as part of their annual operating plan. The cost mostly depends on external integration requirements, platform customizations, data backup and availability, response times required in the SLA, etc.

At this point, we can give you an estimate for the running cost for 1 organisation/center of ~5.000 EUR / yearly software license and support. Additional cost would be the ticket procurement and potential maintenance visits on location (before/after season).

7.4 EU Funding: Alpine Space Interreg

The expansion of the fixed-station network across the DNS region: and beyond into neighbouring Alpine Nordic destinations in Austria, Trentino, and Switzerland: represents a strong candidate for co-funding under the Alpine Space Interreg programme (2028+ cycle). PRIOT has active experience in EU-funded R&D projects (EU i3 Programme, ESA/CASSINI ecosystem) and is available to act as technical partner or co-applicant in an Interreg application. A joint

submission positioning DNS as the first intelligent Nordic ticketing network in the Alps would align directly with EU digitalisation and sustainable tourism funding priorities.

8. Selected References

Planica Nordic Centre: Real-Time Access Control & Visitor Analytics | 2022–present

Why it matters: same technology, same domain, already in production.

PRIOT Smart Track has been running at the Planica Nordic Centre since 2022, managing daily and seasonal passes across the most-used Nordic ski facility in Slovenia. The system handles RFID-based ticket validation, seasonal and daily pass management, personal timing via QR codes, and track-occupancy visualisation for administrators. In its first operational season, the platform recorded over 4,100 laps and 580+ kilometres tracked. This is not a proof of concept: it is a proven, operational system in exactly the environment DNS is targeting.

Telespor / Nortura: Nordic IoT Tracking Platforms | 2023–present

Why it matters: demonstrates PRIOT's ability to scale IoT infrastructure to national and cross-border scope.

PRIOT rebuilt Telespor's user portal for real-time livestock tracking used by farmers across Norway and Sweden, integrating directly with GPS hardware from IRNAS. For Nortura: Norway's largest meat and egg cooperative: PRIOT built a national-scale connected animal platform handling BLE ear tags and live location data from thousands of farms, designed to scale to millions of connected devices. The same architectural principles (cloud-first, real-time events, offline-capable field devices, multi-region data scoping) apply directly to the DNS Smart Track network rollout.

GIC: Betonarna 2.0 Concrete Laboratory Platform | 2023–2024

Why it matters: GoDigital 2025 award-winning IoT digitalisation: shows PRIOT's ability to eliminate legacy paper-based workflows with robust, data-driven systems.

PRIOT digitised the full concrete quality testing process for GIC d.o.o.: from sample registration through automated IoT sensor data capture to digital reporting and full traceability. The project eliminated paper workflows entirely, reduced raw material consumption by 20–30%, and won the GoDigital 2025 Special Award for Innovative Digital Transformation of Industry (GZS + ICT Association of Slovenia). It illustrates PRIOT's model: take a domain with fragmented, manual, or paper-based operations and replace it with an integrated data platform: exactly the transformation DNS is pursuing.

FruitRace: Smart Fruit Production Platform (EU-funded) | 2024–2025

Why it matters: demonstrates PRIOT's successful delivery of an EU grant-funded IoT project: directly relevant to the Interreg ambition.

Under EU i3 Programme grant agreement No 101083989, PRIOT developed an IoT and analytics platform for fruit beverage manufacturing covering real-time production monitoring, full ingredient traceability, and HACCP/IFS compliance automation. The project involved multiple consortium partners across public and private organisations. PRIOT's experience managing EU-funded technical delivery: from grant application to working production system: makes it a credible co-applicant for an Alpine Space Interreg proposal for the DNS fixed-station network.

9. R&D Activities

PRIOT maintains an active research and development programme led by its two founding directors, both of whom hold PhDs from the Faculty of Electrical Engineering at the University of Ljubljana. This R&D culture is not incidental: it is the reason PRIOT's systems are built to a different standard than typical integration shops, and it is what makes PRIOT a credible partner for a long-term, evolving programme like the DNS Smart Track network.

Grega Logar, PhD: doctoral dissertation: *"Modeling and evaluation of ski jumping using data from a wearable measurement system"*. Research areas: wearable IMU sensing, biomechanical motion capture, sports analytics. PRIOT's presence in Nordic sports technology is not opportunistic: it is grounded in years of applied research in exactly this domain. The Planica system was built by people who understand how skiers move, how Nordic venues operate, and what data is

meaningful in this environment.

Blaž Jakopin, PhD: doctoral dissertation: *"Analysis and control of a variable stiffness actuator"*. Research areas: robotics, compliant actuation, human-robot interaction, industrial IoT and digital twin architectures.

Key R&D areas relevant to DNS:

- **Industrial IoT & Digital Twin:** real-time sensor networks, edge devices, cloud data pipelines, digital replicas of physical infrastructure
- **Wearable Sensing & Sports Analytics:** motion capture, IMU-based tracking, biomechanical modelling: the research foundation of the Smart Track system
- **Satellite Remote Sensing & AI:** deep learning on multispectral satellite time-series data for environmental monitoring, relevant to the DNS long-term vision of smart trail management (snow conditions, climate adaptation)
- **Lab Automation & Smart Platforms:** integration of complex hardware with cloud-based platforms, experience from pharmaceutical (Novartis) and construction (GIC) environments

Awards & recognition:

- GoDigital 2025: TOP 3 + Special Award for Innovative Digital Transformation of Industry (GZS + ICT Association of Slovenia)
- CASSINI Hackathon Winner 2022: European space data application challenge
- EUSPA myEUSpace Competition Winner 2022: European Union Agency for the Space Programme

EU & public funding track record: EU i3 Programme participant (FruitRace, grant No 101083989); ESA BIC / CASSINI ecosystem participant; experience preparing and executing grant-funded R&D projects with multi-partner consortia.